Sigmoid function, $f(x) = \frac{1}{1+e^{-x}}$

Let, $y = f(x)$

So, $y = \frac{1}{1+e^{-x}}$

1st derivative:

$$y = (1+e^{-x})^{-1}$$

$$\frac{dy}{dx} = -1 \cdot (1+e^{-x})^{-2} \cdot \frac{d}{dx}(1+e^{-x})$$

Now, $\frac{d}{dx}(1+e^{-x}) = -e^{-x}$

So, $\frac{dy}{dx} = -(1+e^{-x})^{-2}(-e^{-x})$

$$\frac{dy}{dx} = \frac{e^{-x}}{(1+e^{-x})^2}$$

since, $y = \frac{1}{1+e^{-x}}$

$$1-y = 1 - \frac{1}{1+e^{-x}}$$

$$1-y = \frac{e^{-x}}{1+e^{-x}}$$

so, $y(1-y) = \frac{1}{1+e^{-x}} \cdot \frac{e^{-x}}{1+e^{-x}}$

$$= \frac{e^{-x}}{(1+e^{-x})^2}$$

$$y' = y(1-y)$$

$$\therefore f'(x) = f(x)(1-f(x))$$